Policy Optimization: Actor-Critic to PPO

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Policy Optimization: **Recap (Day 3)**

Recap: What Day 3 Established

- **The objective is unchanged from Day 3** — maximize the expected discounted return:

$$\mathcal{J}(\theta) = \mathbb{E}\left[\sum_{t\geq 0} \gamma^t r_t \mid \pi_\theta\right], \qquad \theta^\star = \arg\max_\theta \mathcal{J}(\theta)$$

- Day 3 derived its gradient and, after variance reduction, ended at the estimator:

$$\nabla_\theta \mathcal{J}(\theta) \approx \sum_{t\geq 0} A^\pi(s_t, a_t)\, \nabla_\theta \log \pi_\theta(a_t|s_t)$$

- $A^\pi(s,a) = Q^\pi(s,a) - V^\pi(s)$ is the **advantage function** — the best baseline
- Using A^π instead of raw returns reduces variance without introducing bias
- **Unresolved question (Day 3's closing slide):** how do we actually estimate A^π?

Recap: What Today Delivers

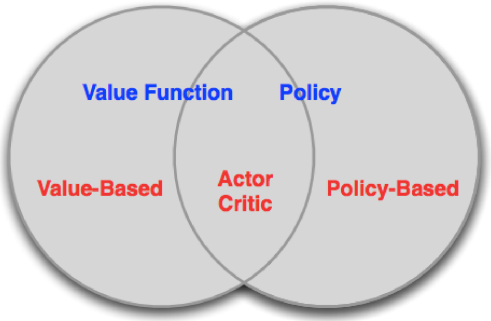
- **Today we cash that promise.**
- We learn a critic V_ϕ alongside the policy — this is the **Actor-Critic** architecture
- Good news: we only need to learn V^π, not Q^π — one network, one output per state
- The critic is trained by **TD learning** (which we unpack in a few slides); its one-step error is what gives us an online, low-variance estimate of A^π
- Then we scale up: A2C/A3C $\rightarrow$ trust region (TRPO) $\rightarrow$ clipped objective (PPO)

Learning Outcomes

- Explain **TD learning** — what the TD error is, why the critic is trained on its *square*, and how it connects to the Q-learning update from Days 1–2
- Explain why a learned critic reduces variance and enables online updates (**Actor-Critic**), and derive the TD-residual estimate of the advantage
- Use λ (**GAE**) to trade bias against variance, and read $\hat{A}_t$ as "how much better than usual this action turned out"
- Describe **A2C** and A3C at a high level and identify when each is preferred
- Explain why we need a **surrogate objective** $\mathcal{L}(\theta) = \mathbb{E}_t[r_t(\theta)\hat{A}_t]$ to score a candidate policy from already-collected data — and why it is only trustworthy nearby
- State the **TRPO** trust-region constraint, and why its cost is what motivates PPO
- Implement the **PPO** clipped objective, and identify the three terms of its combined loss

Policy Optimization: **Actor-Critic Methods**

The Two Worlds of RL



- **Value-based** methods (Days 1–2): learn V or Q, derive policy implicitly
- **Policy-based** methods (Day 3): learn π_θ directly, no value function
- **Actor-Critic** sits at the intersection: learn *both* a policy *and* a value function

From Baselines to Critics

- Day 3: subtracting a **state-dependent baseline** $b(s)$ from the reward-to-go reduces variance with zero bias:

$$\nabla_\theta J(\theta) \approx \sum_{t \geq 0} \Big(\underbrace{\textstyle\sum_{t' \geq t} \gamma^{t'-t} r_{t'}}_{\text{reward-to-go}} - b(s_t) \Big) \nabla_\theta \log \pi_\theta(a_t|s_t)$$

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- The reward-to-go is one *sampled* rollout starting from (s_t, a_t). Its *expectation* is, by definition, the Q-value: $Q^\pi(s_t, a_t) = \mathbb{E}\Big[\sum_{t' \geq t} \gamma^{t'-t} r_{t'} \mid s_t, a_t, \pi\Big]$
- So the same estimator, written in expectation, is:

$$\nabla_\theta J(\theta) = \mathbb{E}_t\big[\big(Q^\pi(s_t, a_t) - b(s_t)\big) \nabla_\theta \log \pi_\theta(a_t|s_t)\big]$$

- Best choice of baseline: $b(s) = V^\pi(s)$, giving the advantage $A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$

From Baselines to Critics

- So the ideal weight on each $\nabla_\theta \log \pi_\theta(a_t|s_t)$ is $A^\pi(s_t, a_t) = Q^\pi(s_t, a_t) - V^\pi(s_t)$
- But we don't know V^π — we have to **learn it**
- This turns the baseline into a **critic**: a second network $V_\phi(s)$ trained alongside the actor π_θ
- **Actor** $\pi_\theta(a|s)$: selects actions, updated by the policy gradient (Day 3)
- **Critic** $V_\phi(s)$: evaluates states, updated by **TD learning** — the subject of the next slide

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- **Critic** $V_\phi(s)$: evaluates states, updated by **TD learning** — the subject of the next slide
- **Why only V, and not Q as well?** $A^\pi = Q^\pi - V^\pi$ looks like it needs both. It doesn't: one observed reward plus the critic at the next state already stands in for Q^π. That is exactly what the TD error does.

What Is TD Learning? (You Have Already Used It)

Temporal Difference (TD) learning = update a prediction using a *later, better-informed* prediction, instead of waiting for the final outcome.

- **Monte Carlo (Day 3):** to learn "how good is s_t?", play the episode to the end and use the actual **reward-to-go** $R_t = \sum_{t' \geq t} \gamma^{t'-t} r_{t'}$ — the same quantity REINFORCE used. Correct on average, but you wait, and the answer swings wildly.
- **TD:** take *one* step. You now know the real reward r_t and you can look up your own estimate of the next state. That combination is already a better estimate of s_t's value than your current guess:

$$\underbrace{V_\phi(s_t)}_{\text{old guess}} \quad \text{vs.} \quad \underbrace{r_t + \gamma V_\phi(s_{t+1})}_{\text{TD target: 1 real reward + guess for the rest}}$$

- The gap between them is the **TD error**:

$$\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$$

- Using your own later estimate inside the target is called **bootstrapping**.

TD Learning: Same Rule You Saw in Days 1–2

- Every value-based update in this course has been a TD update — we simply had not named it:

$$\textbf{Q-learning (Day 1):} \quad Q(s,a) \leftarrow Q(s,a) + \alpha\big[\, r + \gamma \max_{a'} Q(s',a') - Q(s,a) \,\big]$$

$$\textbf{SARSA (Day 2):} \quad Q(s,a) \leftarrow Q(s,a) + \alpha\big[\, r + \gamma Q(s',a') - Q(s,a) \,\big]$$

$$\textbf{TD for our critic:} \quad V_\phi(s_t) \leftarrow V_\phi(s_t) + \alpha\big[\, \underbrace{r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)}_{\delta_t} \,\big]$$

- Same shape every time: *move the prediction a little toward the target*. Day 1 called the bracket the **Bellman error**; for a one-step sample it is the TD error.

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- Same shape every time: *move the prediction a little toward the target*. Day 1 called the bracket the **Bellman error**; for a one-step sample it is the TD error.
- **Why TD suits an actor-critic:** it gives a learning signal at *every* timestep (no waiting for the episode to end), and it works in continuing tasks with no terminal state.
- **The catch:** the target contains our own guess $V_\phi(s_{t+1})$. If the critic is wrong, we are chasing a wrong target — TD is **biased** but **low variance**, the mirror image of Monte Carlo.

Training the Critic: Why the *Squared* Error?

The critic is trained by gradient descent on the *squared* TD error — Day 2's DQN loss $\left(y_i - Q\right)^2$, applied to V:

$$\mathcal{L}(\phi) = \mathbb{E}_t\left[\delta_t^2\right] = \mathbb{E}_t\left[\left(\underbrace{r_t + \gamma V_\phi(s_{t+1})}_{\text{target } y_t} - V_\phi(s_t)\right)^2\right]$$

- **Why not just δ_t?** It is signed — over-estimates give $\delta_t < 0$, under-estimates $\delta_t > 0$ — so over a batch they *cancel*: a critic wrong in both directions would report a loss of zero.

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- **Why not $|\delta_t|$?** Non-negative too, but its gradient has constant magnitude (a tiny error and a huge one pull equally hard) and is undefined at 0. Squaring gives $\nabla_\phi \mathcal{L} = -2\,\mathbb{E}[\delta_t \nabla_\phi V_\phi(s_t)]$: **the correction scales with the size of the mistake** — the "nudge by $\alpha\,\delta_t$" rule above.

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- **The deeper reason:** $V^\pi(s)$ *is* an expectation, and the minimizer of a squared error is exactly the mean of the target — squared error is the loss whose optimum is the quantity we want. ($|\delta_t|$ would converge to the *median* return.)

The TD Residual as Advantage

- We introduced δ_t as the critic's *training* signal. Here is the second, less obvious role it plays — the one that makes actor-critic work:

$$\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t) \qquad \text{(the } \textbf{TD residual}\text{)}$$

- When $V_\phi = V^\pi$ (a perfect critic), the **expected** TD residual equals the advantage:

$$\begin{aligned}\mathbb{E}[\delta_t \mid s_t, a_t] &= r(s_t, a_t) + \gamma \mathbb{E}_{s_{t+1}}[V^\pi(s_{t+1})] - V^\pi(s_t) \\ &= Q^\pi(s_t, a_t) - V^\pi(s_t) \;=\; A^\pi(s_t, a_t)\end{aligned}$$

where $r(s_t, a_t) = \mathbb{E}[r_t \mid s_t, a_t]$ is the expected reward function. In practice we use the observed r_t and a learned V_ϕ, so δ_t is an approximation of A^π.

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- So we can use δ_t as a **low-variance, online** estimate of A^π — no full trajectory needed
- This is what Day 3 deferred: we now have a way to estimate A^π at every timestep
- **One number, two jobs:** δ_t trains the critic (via $\mathcal{L}(\phi) = \mathbb{E}_t[\delta_t^2]$) *and* weights the actor's gradient. That is the whole actor-critic loop.

Actor-Critic Algorithm

Initialise actor θ, critic ϕ;
for *each episode* **do**
  observe initial state s_0;
  for $t = 0, 1, 2, \ldots$ **do**
    sample $a_t \sim \pi_\theta(\cdot|s_t)$;
    take a_t, observe r_t, s_{t+1};
    $\delta_t \leftarrow r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$;
    $\theta \leftarrow \theta + \alpha_\theta \nabla_\theta \log \pi_\theta(a_t|s_t)\, \delta_t$;
    $\phi \leftarrow \phi - \alpha_\phi \delta_t \nabla_\phi V_\phi(s_t)$;
  end
end

Algorithm: One-step Actor-Critic (online)

- **Actor** steps along the policy gradient, weighted by δ_t
- **Critic** regresses $V_\phi(s_t)$ toward $r_t + \gamma V_\phi(s_{t+1})$
- The same δ_t drives *both* updates — no full trajectory needed
- Note the signs: the actor *ascends* $(+)$ to maximize return; the critic *descends* $(-)$ to minimize $\mathbb{E}[\delta_t^2]$
- This is the pure **1-step TD** form. It is complete and it works — but the single δ_t is a crude advantage estimate, which is what we fix next

Why Actor-Critic Helps

- **Variance reduction vs REINFORCE:** the TD residual δ_t is a much lower-variance signal than REINFORCE's reward-to-go minus baseline, $R_t - b(s_t)$, because it bootstraps off V_ϕ rather than waiting for the full trajectory
- **Online updates:** no need to wait until the end of an episode — the critic provides a training signal at every timestep
- **Bootstrapping:** propagating value estimates forward allows the agent to learn from partial trajectories and even in continuing (non-episodic) tasks

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- **Bootstrapping:** propagating value estimates forward allows the agent to learn from partial trajectories and even in continuing (non-episodic) tasks
- **Trade-off:** if V_ϕ is a bad critic, the TD residual is a biased advantage estimate — the actor is being guided by a noisy critic
- Both networks must be trained together carefully; instability can arise if the critic lags the actor

$\Rightarrow$ So we have swapped REINFORCE's *variance* problem for a *bias* problem. Neither extreme is right — the next section builds the dial between them.

Policy Optimization: **Better Advantage Estimates**

The Question We Have Been Dodging: How Far Do We Look Ahead?

The actor's update is only as good as the number $\hat{A}_t$ we multiply it by. We have met **two** ways to produce that number, and they are opposite extremes:

Monte Carlo (Day 3)

$$\hat{A}_t = \underbrace{R_t}_{\text{all real rewards}} - V_\phi(s_t)$$

$R_t = \sum_{t' \geq t} \gamma^{t'-t} r_{t'}$ (reward-to-go)

Never trusts the critic beyond s_t.
$\Rightarrow$ **unbiased**, but every random event in the rest of the episode is baked into the number: **huge variance**.

1-step TD (this lecture)

$$\hat{A}_t = \delta_t = \underbrace{r_t}_{\text{1 real reward}} + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$$

Trusts the critic completely after one step.
$\Rightarrow$ **low variance**, but inherits every mistake V_ϕ makes: **biased**.

- Neither is obviously right — and *which* is better changes **during training**: a freshly initialised critic is worthless (trust the real rewards), a well-trained one is excellent (trust the critic, keep the variance low).

Why This Matters in Practice

- **Concrete failure of 1-step TD:** a robot knocks over a cup at step t, but the penalty only arrives at step $t+30$. The 1-step residual δ_t can only see that penalty *through* $V_\phi(s_{t+1})$. If the critic has not learned about the delayed penalty yet, $\delta_t \approx 0$ — the actor is told the disastrous action was fine.

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- **Concrete failure of Monte Carlo:** the same robot later gets a random $+100$ from an unrelated event at step $t+200$. That windfall is added to $\hat{A}_t$ for *every* earlier action, good or bad — pure noise drowning the real signal.

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$\Rightarrow$ The real knob is: **how many real rewards do we look at before handing off to the critic?**

- One step is one answer. The whole episode is another. **Everything in between is also allowed** — and that family is what the next slide builds.
- **Generalized Advantage Estimation (GAE)** is the standard way to navigate this family. It is not an exotic add-on: PPO, A2C, and nearly every modern on-policy implementation use it, and you will use it in the lab.

From n-step Returns to GAE

- Between 1-step TD and full Monte Carlo lies a whole family of **n-step advantage estimators**:

$$\hat{A}_t^{(n)} = \underbrace{r_t + \gamma r_{t+1} + \cdots + \gamma^{n-1} r_{t+n-1} + \gamma^n V_\phi(s_{t+n})}_{n\text{-step return}} - V_\phi(s_t)$$

  - Take n real rewards, then hand off to the critic. Small n: low variance, high bias; large n: high variance, low bias
  - $n = 1$ recovers δ_t; $n = \infty$ recovers Monte Carlo — our two extremes are just members of this family

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- Each $\hat{A}_t^{(n)}$ telescopes into a sum of TD residuals we already compute:

$$\hat{A}_t^{(n)} = \sum_{l=0}^{n-1} \gamma^l \, \delta_{t+l}$$

- **So which n?** Any single choice is a hard cutoff — full trust in real rewards up to step n, then abrupt total trust in the critic — and it is one more hyperparameter to tune per task.

GAE: Don't Pick n — Average Over All of Them

- **Generalized Advantage Estimation** takes a **weighted average of every** $\hat{A}_t^{(n)}$, with weights decaying geometrically at rate $\lambda \in [0, 1]$ so shorter (lower-variance) estimates count most:

$$\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} = (1-\lambda)\sum_{n=1}^{\infty} \lambda^{n-1}\, \hat{A}_t^{(n)} \;=\; \sum_{l=0}^{\infty} (\gamma\lambda)^l\, \delta_{t+l}$$

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- **Why an average helps:** no cliff at step n — the handover from real rewards to the critic is gradual, and no single n has to be right.
- **Why it is free:** the average looks expensive but — via the telescoping on the previous slide — collapses into **one discounted sum of the TD residuals we already have**. Same cost as 1-step TD.
- λ smoothly tunes the bias–variance trade-off (next slide: the limits)

GAE: The Two Limits

- The single knob λ in $\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} = \sum_{l\geq 0}(\gamma\lambda)^l\,\delta_{t+l}$ interpolates the bias–variance trade-off:
- $\lambda = 0$: pure 1-step TD — low variance, biased
- $\lambda = 1$: full Monte Carlo ($R_t - V_\phi(s_t)$) — unbiased, high variance
- $\lambda \approx 0.95$: standard in PPO implementations
- **Read λ as "how much do I trust my critic?"** — small λ leans on V_ϕ, large λ leans on the observed rewards

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- $\lambda \approx 0.95$: standard in PPO implementations
- **Read λ as "how much do I trust my critic?"** — small λ leans on V_ϕ, large λ leans on the observed rewards
- **Computed backwards in one pass** over a rollout — the form you will implement:

$$\hat{A}_t = \delta_t + \gamma\lambda\,\hat{A}_{t+1}, \qquad \hat{A}_T = 0$$

- We write $\hat{A}_t$ for short: our **practical estimate** of the true advantage $A^\pi(s_t, a_t)$

A2C: Synchronous Advantage Actor-Critic

- **Key idea:** run N parallel actors sharing the same θ and ϕ
- All actors step simultaneously; gradients are aggregated and a single synchronous update is applied
- The shared critic sees diverse experience from all environments — more stable training signal
- **Policy loss:** $\mathcal{L}^{\pi}(\theta) = -\mathbb{E}_t\left[\log \pi_\theta(a_t|s_t)\,\hat{A}_t\right]$
- **Value loss:** $\mathcal{L}^{V}(\phi) = \mathbb{E}_t\left[\delta_t^2\right]$
- A2C is what most modern Actor-Critic implementations actually use
- Parallel actors de-correlate experience without a replay buffer (on-policy)

A3C: Asynchronous Variant

- **A3C** (Mnih et al., 2016): Asynchronous Advantage Actor-Critic
- Workers run asynchronously on CPU threads, applying gradients to a shared model without locks (Hogwild-style updates)
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- A3C's asynchrony introduces **gradient staleness**: a worker may apply a gradient computed under an older version of θ

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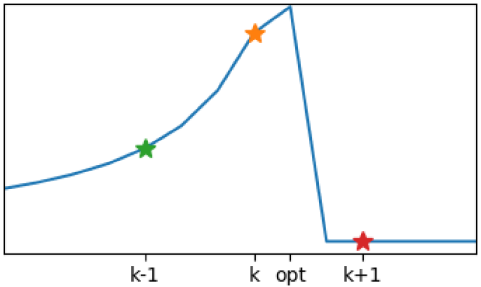
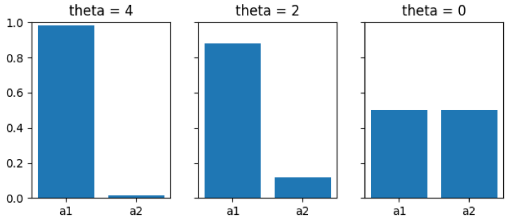
- A3C's asynchrony introduces **gradient staleness**: a worker may apply a gradient computed under an older version of θ
- **Going forward:** we use A2C as the basis for PPO; A3C is historical context
- But parallel actors only *accelerate* data collection — they do not fix the underlying issue that **a single gradient step can still destroy the policy**. That motivates the next section.

Why Step Size Matters





- A large gradient step in *parameter space* can cause a catastrophically large shift in *policy space*
- Once a policy collapses (e.g., always picks one bad action), the agent collects bad data and the policy is hard to recover
- We need to control how much the policy distribution changes, not just how much θ changes

What We Actually Need Before We Can Take a Safe Step

- Last slide's conclusion: *don't let the policy move too far in one update*. But to act on that we first need to answer a prior question:

  Given a candidate π_θ, how much better or worse is it than $\pi_{\theta_{\text{old}}}$ — without running it?

- Why we can't just run it: evaluating $\mathcal{J}(\theta)$ means collecting a fresh batch of rollouts for *every* candidate θ we consider. That is prohibitively expensive — and if the candidate is bad, we only find out by damaging the agent.

$\Rightarrow$ So we need a **score we can compute from the batch we already collected** — a stand-in for $\mathcal{J}(\theta)$, called a **surrogate objective**. Next slide builds it from scratch.

Building That Score Out of What the Batch Already Contains

What the batch contains: for every timestep, the state s_t, the action a_t we took, and $\hat{A}_t$ — whether that action turned out better $(+)$ or worse $(-)$ than usual.

- A candidate π_θ cannot change what happened. The *only* thing it changes is **how likely each of those actions would have been**.

[1] $r_t(\theta)$ is the probability ratio, not the reward r_t from earlier lectures — distinguished by the θ argument.

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- Measure that per action with the **probability ratio**:[1]

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$$

1 : would take a_t just as often
2 : twice as often 0.5 : half as often

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- Score each sample by **(how much more the new policy wants it) $\times$ (how good it was)**, and average:

$$\mathcal{L}(\theta) = \mathbb{E}_t\Big[\, r_t(\theta)\, \hat{A}_t \,\Big]$$

[1] $r_t(\theta)$ is the probability ratio, not the reward r_t from earlier lectures — distinguished by the θ argument.

Why Multiply? A Two-Action Batch

Every column is a number we can just compute: $\hat{A}_t$ from the critic, the probabilities by evaluating each network on (s_t, a_t).

	$\hat{A}_t$	$\pi_{\theta_{\text{old}}}(a_t \mid s_t)$	$\pi_\theta(a_t \mid s_t)$	$r_t(\theta)$	$r_t(\theta)\,\hat{A}_t$
went well	$+2.0$	0.20	0.40	2.0	$+4.0$
went badly	-1.0	0.50	0.40	0.8	-0.8

- $\mathcal{L}(\theta) = \frac{1}{2}(4.0 - 0.8) = +1.6 > 0$: the candidate **doubles down on what worked, backs off what didn't**. Ascending $\mathcal{L}$ searches for exactly that.

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- At $\theta = \theta_{\text{old}}$ every $r_t = 1$, so $\mathcal{L}$ starts at ≈ 0 and its gradient is exactly the Day 3 policy gradient. **This is the objective PPO maximizes.**

Why the Surrogate Only Works Nearby

$\mathcal{L}(\theta)$ is an *approximation* of the true improvement, and it degrades in two ways as π_θ moves away from $\pi_{\theta_{\text{old}}}$:

1. The re-weighting becomes statistically useless

If $\pi_{\theta_{\text{old}}}(a|s)$ is tiny where $\pi_\theta(a|s)$ is large, $r_t(\theta)$ blows up: one rare sample gets an enormous weight and swings the entire gradient. The estimate stays unbiased but its **variance explodes**.

2. We quietly swapped the state distribution

Importance sampling corrects for the *actions*, but we still evaluate them at the states $\pi_{\theta_{\text{old}}}$ happened to visit. A very different policy visits very different states — so the surrogate scores the new policy on a world it will no longer see.

- Both errors are **small when the two policies are close**, and unbounded when they are not.

$\Rightarrow$ This is the principled version of "big steps break things". Everything that follows — TRPO and PPO — is one idea: **maximize** $\mathcal{L}(\theta)$ **while keeping** π_θ **close to** $\pi_{\theta_{\text{old}}}$, differing only in how "close" is enforced.

Policy Optimization: **TRPO**

TRPO: KL Trust Region

TRPO writes "improve, but don't move far" as a **constrained optimization problem**:

$$\underbrace{\max_{\theta} \mathcal{L}(\theta)}_{\text{what we want}} \quad \underbrace{\text{s.t.}}_{\text{"subject to"}} \quad \underbrace{\mathbb{E}_t[D_{KL}(\pi_{\theta_{\text{old}}}(\cdot|s_t) \,\|\, \pi_\theta(\cdot|s_t))] \le \delta}_{\text{the rule the answer must obey}}$$

- **Reading it:** "pick the θ that maximizes the surrogate $\mathcal{L}$, *among only those* θ whose policy stays within δ of the old one." Without the constraint the maximizer is unbounded — $\mathcal{L}$ keeps rewarding a bigger ratio.
- $D_{KL}(p \,\|\, q)$ **(KL divergence)** measures how different two probability distributions are: 0 when identical, growing as they disagree. Here it compares the old and new *action distributions* in the same state s_t — so it measures change in **behaviour**, not in weights.
- δ **is our step-size budget**[2]— a small number we choose (typically 0.01), the radius of the **trust region**. Bigger δ = faster but riskier updates.

[2]Unrelated to the TD residual δ_t from earlier today — unfortunate but standard notation; the KL bound never carries a time subscript.

TRPO: Why Nobody Uses It

- The constraint is on **behaviour** ("the policy must not change much"), but the only thing we can actually turn is the **weights**. Those are not the same thing: some weights barely affect what the agent does, others flip it completely.

- So before every update TRPO has to work out the **conversion rate** between "changed the weights this much" and "changed the behaviour this much" — that conversion table is the **Fisher information matrix**.

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- So before every update TRPO has to work out the **conversion rate** between "changed the weights this much" and "changed the behaviour this much" — that conversion table is the **Fisher information matrix**.
- **The problem:** that table has one entry for every *pair* of weights. A modest 1M-parameter network needs 10^{12} of them — it cannot be built, let alone solved.
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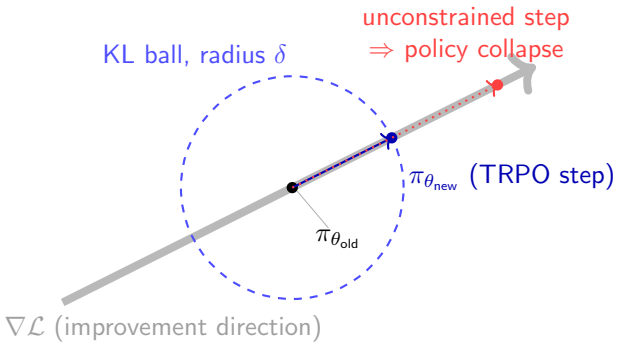
- TRPO has clever machinery to avoid ever building it, but each update still costs several extra passes over the data plus a search to check the constraint was respected.

$\Rightarrow$ **Great theory, painful engineering:** hundreds of lines of second-order code for one policy update. *PPO's question: can we get the same "don't move too far" effect with a one-line change to the loss?*

TRPO: Trust Region in Policy Space



- Each update stays within a **KL ball** around $\pi_{\theta_{\text{old}}}$ (*policy* space, not parameter space): the step climbs $\mathcal{L}$ but is clipped at the boundary, preventing the overshoot that collapses the policy

Policy Optimization: **PPO**

PPO: First-Order Trust Region

- TRPO achieves monotonic improvement but requires second-order optimisation (conjugate gradient, Fisher matrix) — expensive and complex to implement
- **Question:** can we get TRPO-level stability with plain SGD?

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- **Question:** can we get TRPO-level stability with plain SGD?
- **PPO** (Schulman et al., 2017) proposes two variants:
  - **PPO-KL:** add a KL penalty to the objective and adapt the penalty coefficient β over training
  - **PPO-Clip:** clip the probability ratio $r_t(\theta)$ to prevent large policy updates
- PPO-Clip is the standard variant — simpler, no hyperparameter tuning on β, works at least as well empirically

PPO Clipped Objective

- Recall the probability ratio: $r_t(\theta) = \dfrac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$
- The **PPO-Clip objective** is:

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(r_t(\theta)\,\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)\,\hat{A}_t\Big)\Big]$$

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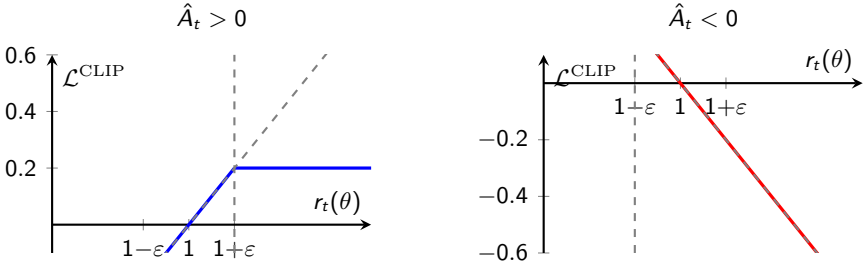
- **Why clip + min?**
  - When $\hat{A}_t > 0$: clipping caps r_t at $1+\varepsilon$, removing the incentive to increase the ratio further
  - When $\hat{A}_t < 0$: clipping floors r_t at $1-\varepsilon$, removing the incentive to decrease the ratio further
  - The **min** ensures the objective is a lower bound — we never benefit from going outside $[1-\varepsilon, 1+\varepsilon]$
- This enforces a soft trust region without any second-order computation

Visualising the Clip



- **Left ($\hat{A}_t > 0$):** benefit grows linearly with r_t until $1+\varepsilon$, then is clipped flat — no reward for pushing the policy further
- **Right ($\hat{A}_t < 0$):** penalty grows as r_t decreases until $1-\varepsilon$, then is clipped flat
- Gray dashed line = unclipped surrogate; colored solid = clipped (what PPO optimises)

PPO Full Algorithm

Algorithm 5 PPO with Clipped Objective

Input: initial policy parameters θ_0, clipping threshold ϵ
for $k = 0, 1, 2, \ldots$ **do**
  Collect set of partial trajectories $\mathcal{D}_k$ on policy $\pi_k = \pi(\theta_k)$
  Estimate advantages $\hat{A}_t^{\pi_k}$ using any advantage estimation algorithm
  Compute policy update

$$\theta_{k+1} = \arg\max_{\theta} \mathcal{L}_{\theta_k}^{CLIP}(\theta)$$

  by taking K steps of minibatch SGD (via Adam), where

$$\mathcal{L}_{\theta_k}^{CLIP}(\theta) = \mathop{\mathrm{E}}_{\tau \sim \pi_k} \left[\sum_{t=0}^{T} \left[\min(r_t(\theta)\hat{A}_t^{\pi_k}, \text{clip}\left(r_t(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_t^{\pi_k}) \right] \right]$$

end for

- **Sample efficiency:** PPO runs multiple SGD epochs over the same rollout — unlike A2C, which discards data after one update
- GAE is used to compute $\hat{A}_t$ once per rollout, before the epoch loop

PPO: The Three Terms of One Loss

Actor and critic share one optimizer step, so their objectives are added into a single number to differentiate — that is all "combined loss" means:

$$\mathcal{L}(\theta, \phi) = \underbrace{\mathcal{L}^{\text{CLIP}}(\theta)}_{\text{1. make good actions likelier}} - c_1 \underbrace{\mathcal{L}^{V}(\phi)}_{\text{2. train the critic}} + c_2 \underbrace{\mathcal{H}(\pi_\theta)}_{\text{3. stay exploratory}}$$

- **1.** $\mathcal{L}^{\text{CLIP}}$ — the clipped surrogate we just built. *Maximized*, so it enters with $+$.
- **2.** $\mathcal{L}^{V} = \mathbb{E}_t[\delta_t^2]$ — the critic's squared TD error. We *minimize* it, hence the minus sign: subtracting it from a quantity we maximize is the same as minimizing it. $c_1 \approx 0.5$ balances the two scales.

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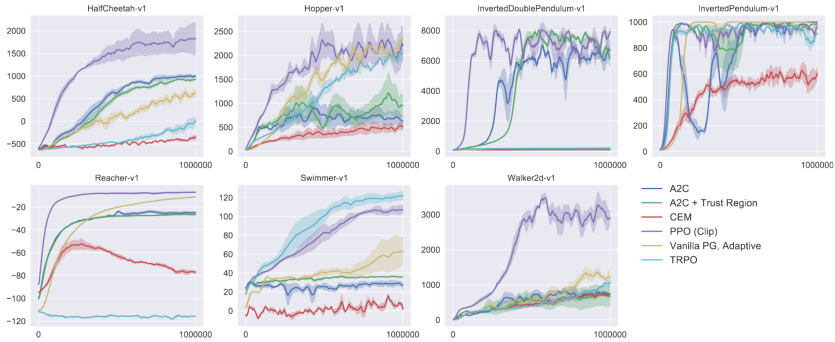
- **3.** $\mathcal{H}(\pi_\theta)$ — the **entropy** of the action distribution: high when the policy is still undecided, near zero when it always picks one action. Adding it *rewards* indecision, which stops the policy from collapsing onto one action before it has explored. c_2 is small (≈ 0.01); the term reappears as a first-class citizen in SAC (Day 6).
- In code this is three lines summed and passed to one `loss.backward()` — no separate actor and critic updates.

PPO in Practice

- The clipped objective is only half the story — these implementation details matter *as much as* ε, and explain why two "correct" PPO implementations can differ by an order of magnitude[3]
- **Advantage normalisation:** standardise $\hat{A}_t$ per minibatch (stabilises gradient scale)
- **Value-loss clipping:** clip the critic update like the policy
- **Gradient clipping:** bound global grad-norm (typically 0.5)
- **KL early stopping:** end the epoch loop once mean KL exceeds a target — a cheap trust-region safeguard
- **Obs / reward normalisation:** running normalisation keeps inputs and returns well-scaled
- **Orthogonal init:** small policy-head gain avoids an over-confident initial policy

[3] Andrychowicz et al. (2020), *What Matters in On-Policy RL?*

PPO Empirical Performance



- PPO matches or exceeds TRPO with far less complexity; outperforms vanilla PG and CEM on most benchmarks
- This is the algorithm you will implement in the lab[4]

[4] Schulman, Wolski, Dhariwal, Radford, Klimov, 2017

Policy Optimization: **GRPO**

PPO Without a Critic: GRPO

- The fragile part of the Actor-Critic loop is the **critic** V_ϕ — training two networks together is the main source of instability
- **Group Relative Policy Optimization** (GRPO; Shao et al., 2024) removes the critic entirely

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- The fragile part of the Actor-Critic loop is the **critic** V_ϕ — training two networks together is the main source of instability
- **Group Relative Policy Optimization** (GRPO; Shao et al., 2024) removes the critic entirely
- Per prompt/state, sample a **group** of G outputs under $\pi_{\theta_{\text{old}}}$, scored with rewards R_i
- Use the **group mean as the baseline** — a Monte Carlo advantage, no value function:

$$\hat{A}_i = \frac{R_i - \overbrace{\text{mean}(R_1, \ldots, R_G)}^{\text{the “Group Relative” baseline}}}{\underbrace{\text{std}(R_1, \ldots, R_G)}_{\text{per-group advantage normalisation}}}$$

GRPO: What Is and Isn't the Baseline

- Only the **mean-subtraction** is the GRPO contribution — that is the "Group Relative" baseline replacing the critic V_ϕ
- The **std-normalisation** is a *separate* stabilisation trick: it is exactly the per-minibatch advantage normalisation from the PPO-in-Practice slide, applied per group — not part of the baseline
- So a Monte Carlo baseline does *not* "need" variance correction; the std term is an orthogonal, optional stabiliser

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- So a Monte Carlo baseline does *not* "need" variance correction; the std term is an orthogonal, optional stabiliser
- Plug $\hat{A}_i$ into the *same* PPO-Clip objective (plus a KL term to a reference policy)
- **Why it matters:** no critic $\Rightarrow$ no two-network instability and $\approx$half the memory
- Since 2024 GRPO is the default for **LLM RL** (DeepSeek-R1, many open recipes) — revisited in **Day 9 (RLHF)**, where the reward is a learned preference model

Policy Optimization: **Summary**

The Algorithm Family Tree

REINFORCE $\rightarrow$ +baseline $\rightarrow$ +critic (A2C) $\rightarrow$ +GAE $\rightarrow$ +trust region (TRPO) $\rightarrow$ +clipping (PPO) $\rightarrow$ $-$critic (GRPO)

Algorithm	**On-policy?**	**Variance**	**Sample eff.**	**Complexity**
REINFORCE	Yes	High	Low	Simple
A2C	Yes	Medium	Medium	Moderate
TRPO	Yes	Medium	Medium	High
PPO	Yes	Medium	Medium	Low
GRPO	Yes	Medium	Medium	Low (no critic)

Summary (1/2): Getting a Good Advantage Estimate

- **TD learning** — update a prediction using a later, better-informed one. The TD error $\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)$ is the same bracket you saw in Q-learning and SARSA; the critic is trained by minimising its *square*
- A **critic** V_ϕ turns Day 3's baseline into something learnable, and δ_t doubles as an online, low-variance estimate of the advantage — one number, two jobs
- **GAE** replaces the crude 1-step δ_t with a λ-weighted average over all lookahead lengths, computed backwards in one pass: $\hat{A}_t = \delta_t + \gamma\lambda\hat{A}_{t+1}$. Read λ as "how much do I trust my critic"
- $\hat{A}_t$ is what *every* algorithm from here on multiplies the log-probability gradient by
- **A2C** parallelises data collection with N synchronous actors; A3C uses asynchronous workers (empirically A2C $\geq$ A3C)

Summary (2/2): Taking a Safe Step

- A large step in *weights* can be a catastrophic step in *behaviour* — and a collapsed policy collects bad data, so it rarely recovers
- To take a step safely we must score a candidate policy **without running it**. The **surrogate objective** $\mathcal{L}(\theta) = \mathbb{E}_t[r_t(\theta)\,\hat{A}_t]$ does this by re-counting already-collected samples by the ratio $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$
- That score is only trustworthy while the two policies stay close — hence every method below is "maximize $\mathcal{L}$, but don't move far"
- **TRPO** enforces this with a hard KL constraint. Correct, but it needs the Fisher matrix: second-order, expensive, painful to implement
- **PPO-Clip** gets the same effect first-order: clip $r_t(\theta)$ to $[1-\varepsilon, 1+\varepsilon]$ and take the min with the unclipped term. One combined loss = clipped objective $- c_1 \times$ critic error $+ c_2 \times$ entropy
- **GRPO** drops the critic entirely, using a group-mean baseline — now the default for LLM RL (DeepSeek-R1)

Limitations of Actor-Critic & PPO

- **Function approximation bias:** the TD residual δ_t is only an unbiased advantage estimate if $V_\phi = V^\pi$. An inaccurate critic introduces bias into the policy gradient
- **Two-network instability:** if the critic lags behind the actor (or vice versa), the feedback loop can diverge — both networks must be trained carefully together (GRPO sidesteps this by dropping the critic entirely)
- **Hyperparameter sensitivity:** PPO performance is sensitive to ε (clip range), λ (GAE), number of epochs per rollout, and entropy coefficient c_2
- **On-policy sample inefficiency:** all methods in this lecture discard experience after each update. In complex environments this can require billions of environment steps
- **Dense-reward assumption:** variance reduction via advantage estimation works well when rewards are frequent. Sparse-reward tasks remain challenging for all on-policy methods

What's Next

- **Key limitation:** on-policy methods discard all experience after each update — data-hungry for complex environments

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- **Day 5 — DDPG & TD3:** off-policy deterministic policies for continuous control — replay buffer, action-value critic $Q_\phi(s, a)$, gradient through $Q_\phi(s, \mu_\theta(s))$
- **Day 6 — SAC:** max-entropy Actor-Critic — the same $c_2\mathcal{H}(\pi_\theta)$ bonus you saw in PPO, now **central** to the objective
- **Day 7 — Exploration:** that entropy bonus *is* exploration via a stochastic policy; Day 7 makes exploration the explicit objective
- **Day 9 — RLHF:** PPO is the algorithm underneath the original InstructGPT/ChatGPT pipeline; GRPO is its critic-free successor

Policy Optimization: **References**

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[2] Sutton, R. S., McAllester, D. A., Singh, S. P., & Mansour, Y. (2000).

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(Performance Difference Lemma — the result behind the surrogate objective; see appendix.)

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Backup — Performance Difference Lemma

- The relative performance identity is the **Performance Difference Lemma**[1]

$$\mathcal{J}(\theta) - \mathcal{J}(\theta_{\text{old}}) = \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t\geq 0}\gamma^t A^{\pi_{\theta_{\text{old}}}}(s_t, a_t)\right]$$

- **Why it matters:** it makes the surrogate $\mathcal{L}(\theta)$ a **principled approximation** to the true objective $\mathcal{J}(\theta)$ — not a heuristic
- When $\pi_\theta \approx \pi_{\theta_{\text{old}}}$, the state distributions induced by the two policies are similar, so samples from $\pi_{\theta_{\text{old}}}$ can stand in for samples from π_θ with *bounded* error — TRPO makes this bound explicit in terms of the maximum KL divergence
- As π_θ drifts away, the bound degrades. This is the theoretical reason for a trust region

[1] Lemma originally due to Kakade & Langford (2002); foundation of TRPO in Schulman et al. (2015).

Backup — Importance Sampling Variance

- The IS-weighted estimator has variance

$$\mathrm{Var}\left[\frac{P(x)}{Q(x)}\, f(x)\right] = \mathbb{E}_{x\sim P}\left[\frac{P(x)}{Q(x)}\, f(x)^2\right] - \mathbb{E}[f]^2$$

- It **explodes** when the ratio $\frac{P}{Q}$ is large where f^2 is large — a single bad sample dominates the estimate
- For **trajectory-level** IS: $\frac{p(\tau|\pi_\theta)}{p(\tau|\pi_{\theta_{\text{old}}})} = \prod_{t=0}^{T} r_t(\theta)$. Small per-step differences **compound multiplicatively** over T steps — the variance is hopeless
- This is why TRPO and PPO use **single-step ratios** and constrain them to stay near 1[2]

[2]Full derivation in OpenAI's Spinning Up: TRPO and Schulman et al. (2015).

Backup — TRPO Algorithm (pseudocode)

Algorithm 3 Trust Region Policy Optimization

Input: initial policy parameters θ_0
for $k = 0, 1, 2, \ldots$ **do**
Collect set of trajectories $\mathcal{D}_k$ on policy $\pi_k = \pi(\theta_k)$
Estimate advantages $\hat{A}_t^{\pi_k}$ using any advantage estimation algorithm
Form sample estimates for

- policy gradient $\hat{g}_k$ (using advantage estimates)
- and KL-divergence Hessian-vector product function $f(v) = \hat{H}_k v$

Use CG with n_{cg} iterations to obtain $x_k \approx \hat{H}_k^{-1} \hat{g}_k$
Estimate proposed step $\Delta_k \approx \sqrt{\frac{2\delta}{x_k^T \hat{H}_k x_k}} x_k$
Perform backtracking line search with exponential decay to obtain final update

$$\theta_{k+1} = \theta_k + \alpha^j \Delta_k$$

end for

- Reference only — students implement PPO, not TRPO, in the lab (see Schulman et al., 2015)